PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Daniel Nwabueze
Program/Batch	B082-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (permission secured) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All footprinting commands were run in Kali Linux (installed as a virtual machine in VirtualBox) and the network scanning was done with Zenmap on a Windows PC. Every step below includes the command used and the result I observed. Screenshots are included as evidence in Section 8, and Section 5 explains why each finding matters.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Kali Linux (in VirtualBox) for reconnaissance and Windows for Zenmap scanning
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts and open ports, and view the network topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** (whois networkwalks.com) to obtain publicly available domain registration information and identify the domain's name servers. The results showed the domain registration and hosting infrastructure.

I then used **WhatWeb** (whatweb networkwalks.com) to identify technologies used by the website. The results identified **WordPress 7.1.4** and **WP Download Manager 3.3.76** running on an Apache web server, along with jQuery 3.7.1, Google Tag Manager and a contact email address (info@networkwalks.com) exposed on the site. HTTP requests were redirected to HTTPS.

Using **Nslookup** (nslookup networkwalks.com), I resolved the domain name to its IP address. The result identified **192.232.216.135**.

I used **Curl** with the -I option (curl -I https://networkwalks.com) to inspect the HTTP response headers. The server answered with HTTP/2 200 and identified itself as Apache. The headers also exposed the WordPress REST API endpoint /wp-json/, and showed that the session cookie is set with the Secure and HttpOnly flags.

Next, I used **Wafw00f** (wafw00f networkwalks.com) to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** (dnsrecon -d networkwalks.com) to enumerate DNS records. The results listed the name servers ns6135.hostgator.com and ns6136.hostgator.com, the mail server mail.networkwalks.com, a Google site verification record, an SPF record ending in ~all (soft fail) and several SRV records for email autodiscovery. Both name servers also revealed their DNS software version (BIND 9.16.23-RH), and the DNSSEC query returned no answer.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my own local network. The practical required me to discover live hosts on the local subnet and generate a network topology.

I entered my local subnet (192.168.0.0/24) as the target in Zenmap and ran a **Quick scan** (nmap -T4 -F 192.168.0.0/24), which checks the 100 most common ports on each host, to identify active hosts.

The scan found three live hosts on the network, in addition to my own computer (localhost):

- Two hosts shown in green in the topology, meaning fewer than 3 open ports were found on each.
- One host shown in yellow, meaning 3 to 6 open ports. Its host details show 3 open ports, 97 closed ports and 100 ports scanned.
- localhost, the computer running Zenmap, shown at the centre of the topology.

Device IP addresses are hidden in this report to protect my network's privacy. Zenmap did not report MAC addresses for these hosts.

After completing the scan, I opened the Topology tab in Zenmap and saved the network topology as a PDF, as required by the practical task. The topology shows my computer connected to the three discovered hosts (see Figure 11 in Section 8).

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers may use exposed version information to look for software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure and software information exposed	DNSRecon showed the BIND version on both name servers; DNSSEC is not enabled	DNS information can help build a broader infrastructure profile	● **Medium**
6	Contact email address visible on the website	WhatWeb found info@networkwalks.com on the page	May be used for phishing or spam	● **Low**
7	Multiple live hosts visible on local network	Zenmap found three live hosts on my home network	Unknown devices on a network should be identified and checked against known devices	● **Low**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records and DNS software exposure**
   DNS records should be checked periodically so that only required information is public. Name servers should avoid revealing software versions, and enabling DNSSEC and a strict SPF policy should be considered.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it adds a useful protective layer in front of the website.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to scan my local network and discover active hosts. I found three live hosts, checked the open ports on one of them, and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

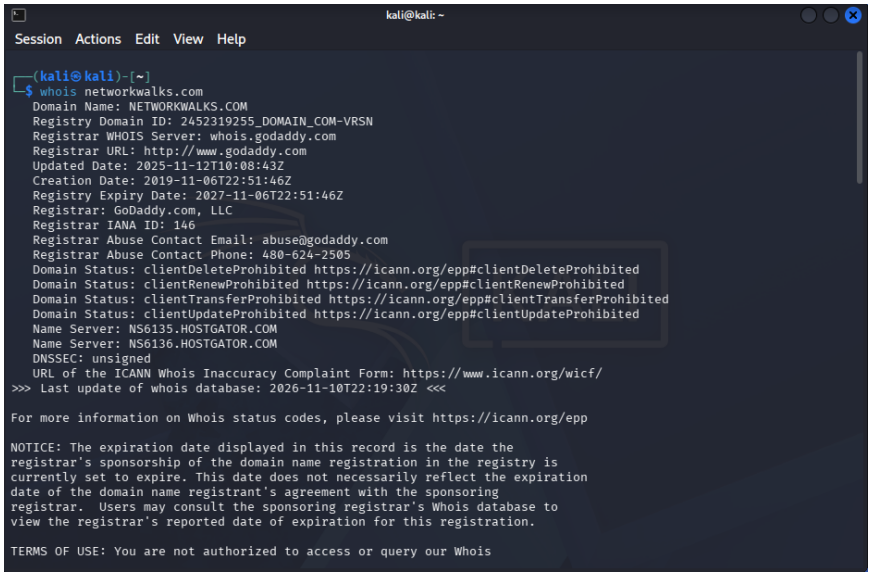


Figure 1: WHOIS lookup of networkwalks.com (part 1 of 4)

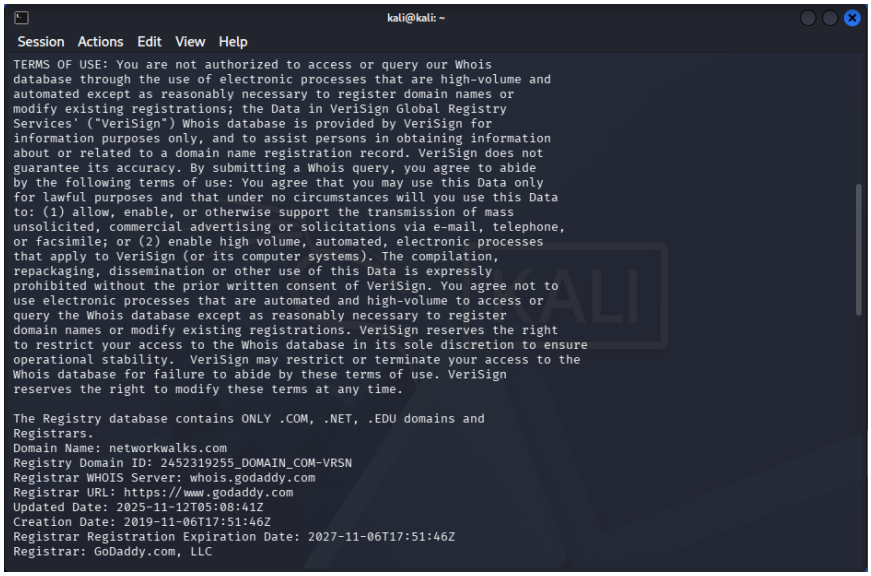


Figure 2: WHOIS lookup (part 2 of 4)

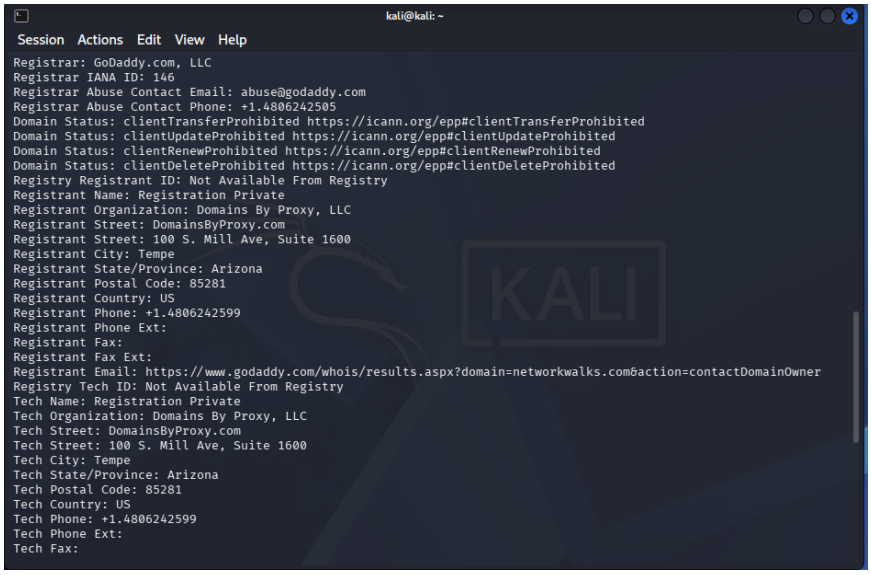


Figure 3: WHOIS lookup (part 3 of 4)

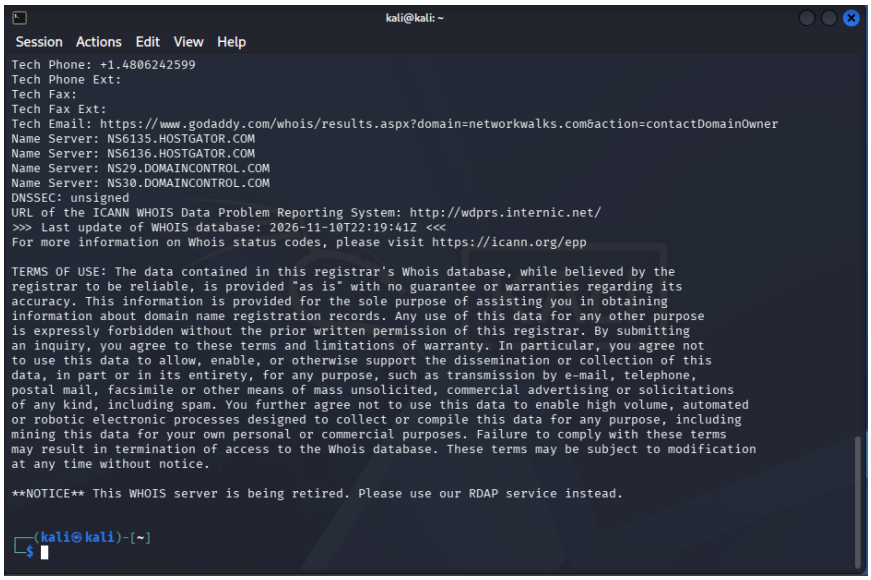


Figure 4: WHOIS lookup (part 4 of 4)

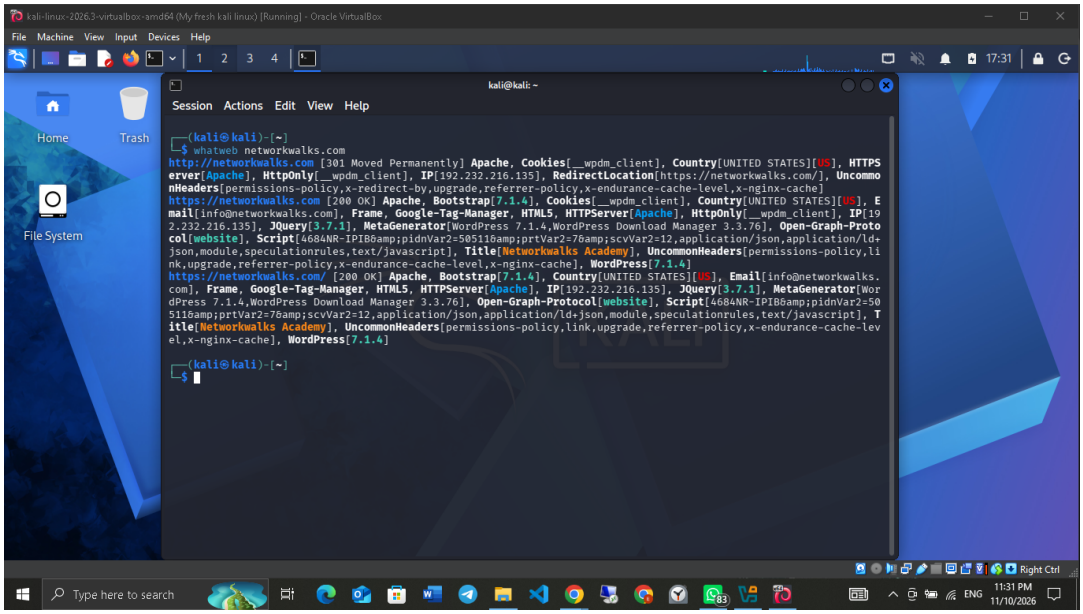


Figure 5: WhatWeb scan of networkwalks.com

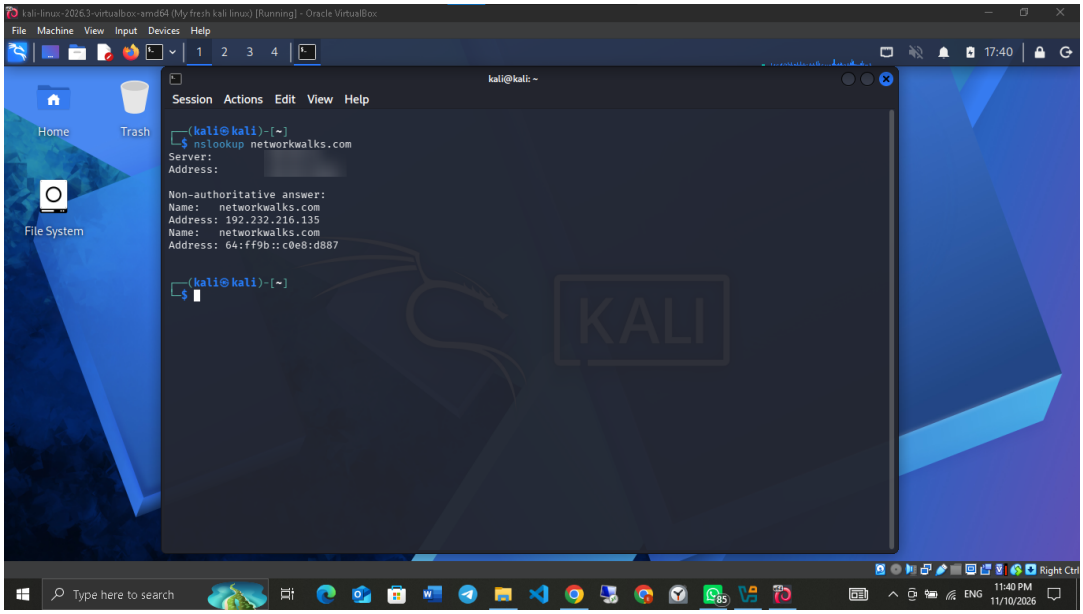


Figure 6: Nslookup of networkwalks.com (local DNS server address hidden)

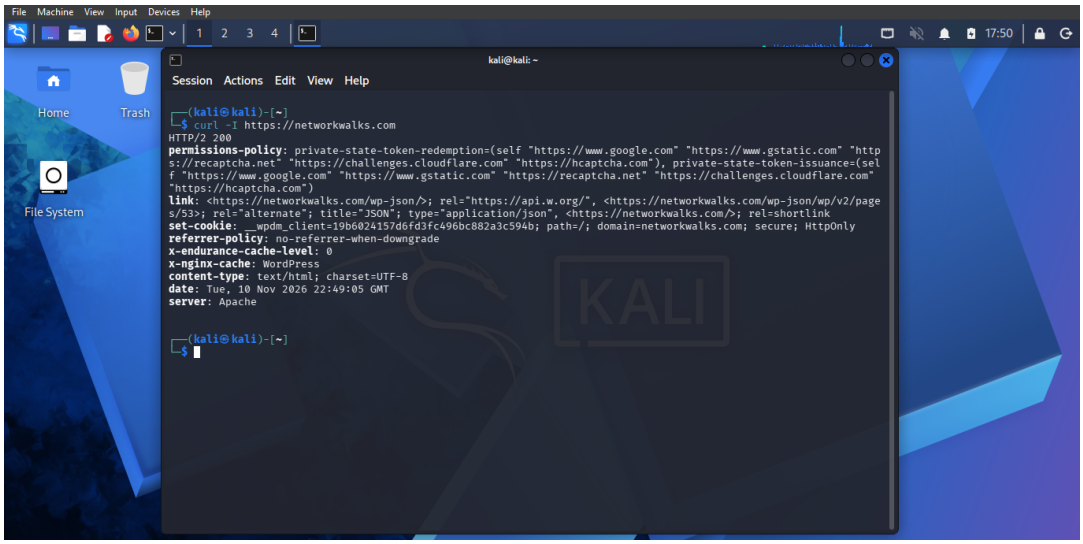


Figure 7: Curl -I HTTP response headers

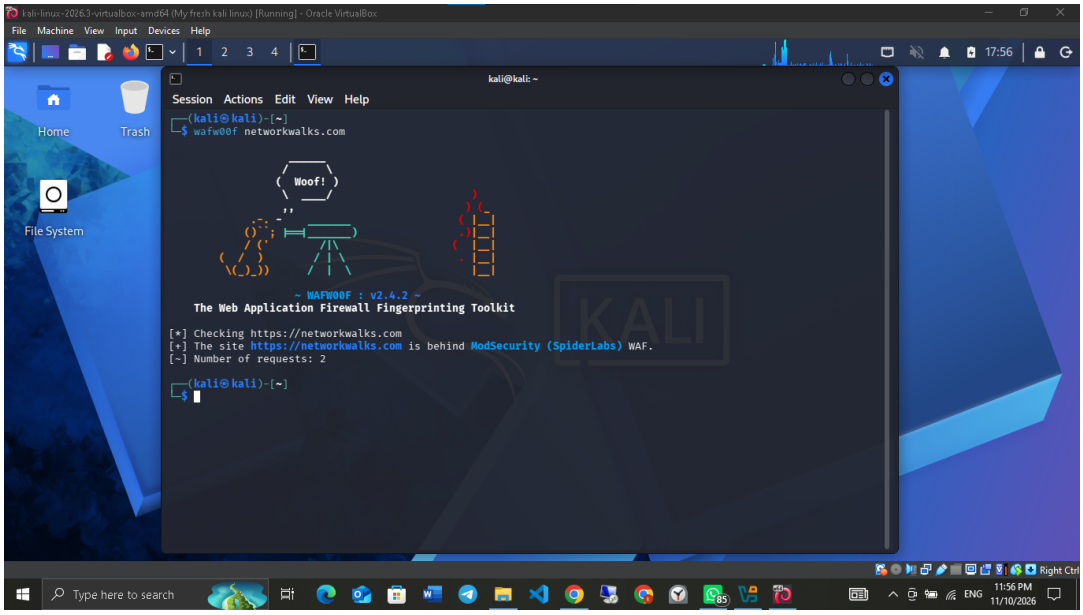


Figure 8: Wafw00f result showing ModSecurity (SpiderLabs)

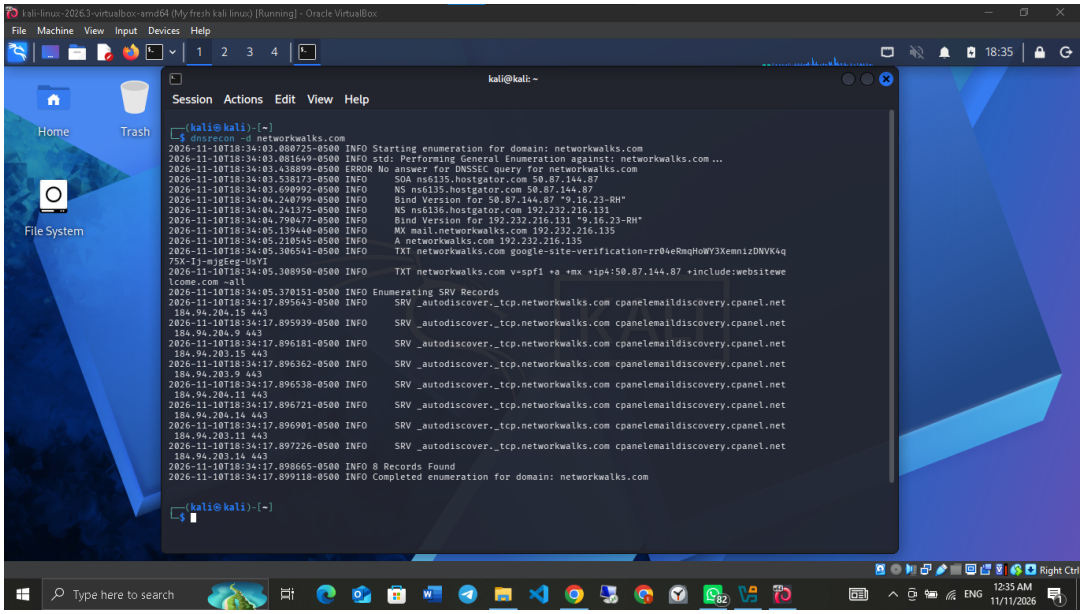


Figure 9: DNSRecon enumeration of networkwalks.com

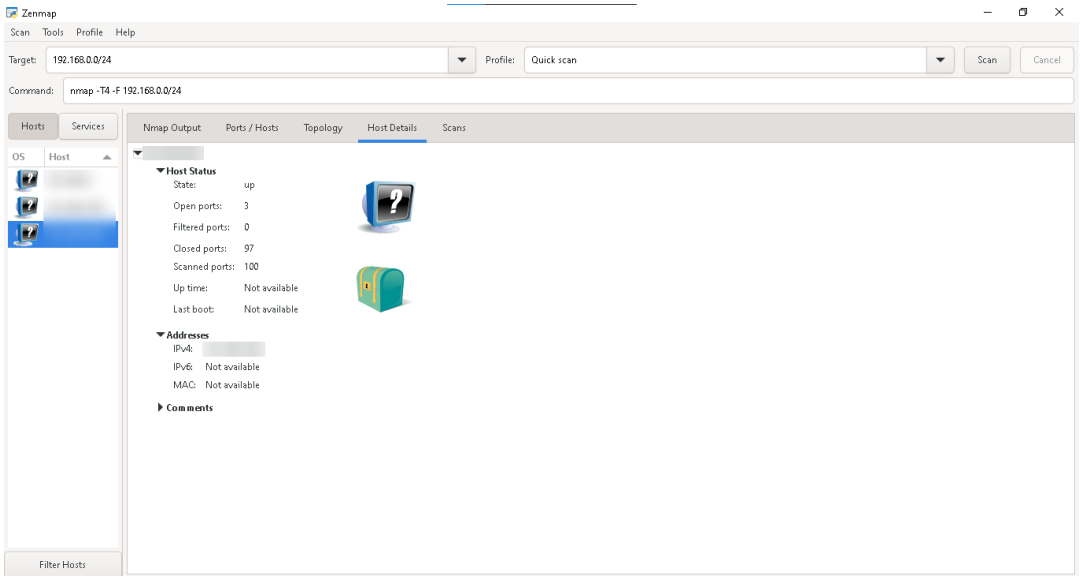


Figure 10: Zenmap Quick scan of my local subnet (device addresses hidden)

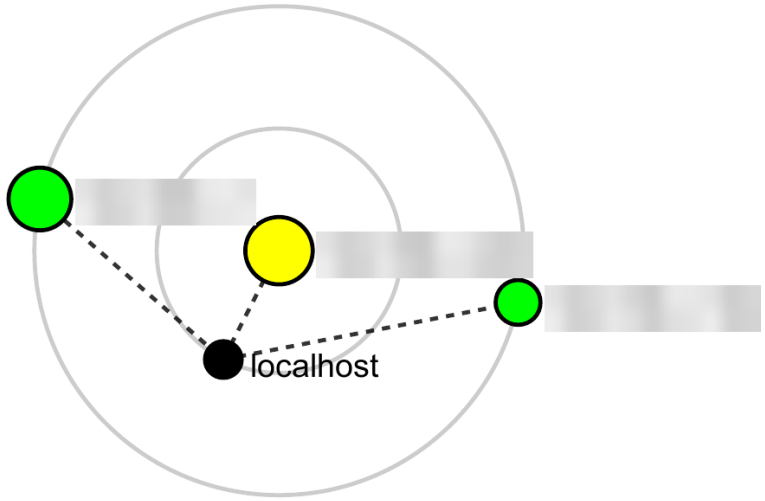


Figure 11: Network topology from Zenmap (device addresses hidden)

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Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** https://github.com/Mr-DannyN/NETWORKWALKS-MRDANNY-B084L-WK2-PM1-PM5-FOOTPRINTING-NETWORK-SCANNING/edit/main/README.md